

Is Einstein Turning In His Grave?

There is a furore in the physics community following an announcement by the Gran Sasso National Laboratory that they have observed neutrinos travelling faster than light. This could open possibilities to send information back in time, time travel and much more



A huge wall of lead 'bricks' used to detect the neutrinos fired from CERN when they arrive at the Gran Sasso laboratory

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It was indeed haunting to see some of the world's greatest scientists waving their hands in frenzy and saying "If this is true, we would have to tear up all our textbooks and rewrite them." There was a lot of excitement too as the new finding meant lots and lots of new research.

Would you believe that a shy little neutrino is at the heart of all this discussion? "Neutrino is an anti-social particle which doesn't like to talk to the world in general. We are crossed by millions of neutrinos every second, but we don't feel them," said Prof. Joao Magueijo of the Imperial College of London, when interviewed for a BBC documentary. The show's anchor put the neutrino more in perspective (or out of it) by saying, "If the atom is the solar system, the neutrino is a

golf ball."

Neutrino is tiny. It is almost weightless, with no electric charge and capable of travelling through solid matter without reacting with anything. In short, it is inconspicuous. However, this miniscule subatomic particle recently created a furore in the physics community as some scientists in Northern Italy observed neutrinos to have travelled faster than light.

If you browse through any physics, computer science, optics, electrical, electronics or communication engineering handbook, or any other encyclopaedia, you will realise that modern science is based on Einstein's theory that the speed of light in vacuum is absolute, and nothing with mass can ever travel faster than light. Whether you recall it by the name of special relativity or the formula

$E=mc^2$, we owe it to Einstein for having unravelled the mysteries of space, light, speed, time and many of the other concepts we take for granted today.

However, an unexpected event has now cast a slight shade of doubt upon Einstein's theories. It is too early to dispute the father of modern physics yet but here is all the information you need to understand this affair as it unfolds in newspapers and journals in the coming months.

One fine day, in the heart of a large mountain...

The Gran Sasso National Laboratory (LNGS) is situated in a national park in Northern Italy, deep within a large mountain. It has a 1400-metre rock cover above it making the lab almost free of cosmic ray flux. The neutron flux is also thousand times less than on the surface, thanks to the very little uranium and thorium content of the dolomite rocks of the mountain. LNGS is the largest underground laboratory in the world for experiments in particle physics, particle astrophysics and nuclear astrophysics with over 750 scientists working on various experiments.

The team that grabbed our attention with its findings is working on the Oscillation Project with Emulsion-tRacking Apparatus (OPERA). The OPERA experiment involves the study of neutrino oscillations. A high-intensity and high-energy beam of muon-neutrinos is produced 730 km away at CERN in Geneva and directed towards the Gran Sasso laboratory. A wall comprising 150,000 bricks of photographic emulsion films interleaved